

# CALVIN MICKELSON

Software Developer

INTERVIEW

RESUME

Fort Collins, CO

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github.com/cabbabbage

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https://cabbabbage.github.io/calvin.github.io/

## SUMMARY

Fort Collins-based software developer with a B.S. in Computer Science and hands-on experience building performance-sensitive C++ systems, custom rendering pipelines, developer tooling, real-time applications, and machine learning/computer vision projects. Strong in C++, Python, and JavaScript, with a focus on fast, debuggable software that works under real constraints.

## TECHNICAL SKILLS

### LANGUAGES

C++, Python, JavaScript, Java, SQL

### FRAMEWORKS/TOOLS

React, Node.js, Git, CMake, TensorFlow, NumPy, Pandas, SDL, MediaPipe

### CORE STRENGTHS

Software engineering, data structures and algorithms, systems design, performance optimization, debugging, developer tooling, rendering pipelines, procedural generation, asset tooling, machine learning

## SELECTED PROJECTS

### ENGINE 2.0 / VIBBLE Engine

C++, SDL, CMake, JSON

- Built a custom 2D engine with dynamic lighting, parallax rendering, layered world composition, and clear separation between runtime, editor, and tooling systems.
- Implemented procedural map generation with room layering, sector-based placement, parent-child room relationships, collision/interaction geometry, and JSON-driven assets.
- Developed editor tooling for asset management, rendering controls, animation caching, and scalable content pipelines supporting rapid iteration.

### auto\_edit

Python, OpenCV, librosa, FFmpeg, EasyOCR

- Built an automated video generator that scans audio/media inputs, detects beats, drop timing, and bass energy, then schedules edits against the music timeline.
- Added OCR-driven text overlay preprocessing, configurable glitch/transition effects, dry-run timeline debugging, and batch rendering of numbered vertical MP4 variants.

### ASL Interpretation Model

Python, TensorFlow, MediaPipe

- Built a real-time ASL recognition pipeline using hand landmark extraction and classification from live video input.
- Trained and evaluated models for sign prediction and implemented preprocessing to improve inference stability and usability.

### SSH\_MAVLINK

Python, Raspberry Pi, MAVLink

- Developed a remote multirotor control system over 4G LTE with real-time telemetry, video streaming, and command/control links.
- Enabled remote manual and autonomous flight workflows using MAVLink-based communication.

LINKS

GitHub

LinkedIn

Portfolio

Email

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## SELECTED PROJECT

### SmileScope

Python, MediaPipe

- Built a real-time facial landmark application to detect smiles and capture annotated screenshots under noisy motion conditions.
- Used MediaPipe landmarks and Python image processing to support lightweight computer vision interaction experiments.

## EXPERIENCE

### Broadcom - Manufacturing Control

Jan 2026 - Mar 2026

Fort Collins, CO

- Supported manufacturing control operations in a high-volume semiconductor production environment.
- Worked across schedule-driven workflows and internal production systems to help maintain continuity and process visibility.

### Collins Control and Electric - Electrician Apprentice

2022 - 2023

Fort Collins, CO

- Assisted with electrical and generator installations and supported troubleshooting of networking, firmware, and control-system issues.

### UNION Bar & Soda Fountain - Line Cook / Prep Cook

2022 - 2025

Fort Collins, CO

- Worked in a fast-paced, high-volume environment while coordinating prep, line execution, and quality standards across daily service.

## EDUCATION

### Colorado State University - B.S. Computer Science

Graduated 2025

Fort Collins, CO

Coursework and project work centered on software engineering, algorithms, systems, data, and applied machine learning.